Probability:

Probability is a branch of mathematics that deals with uncertainty and measures the likelihood or chance of an event occurring.

Probability is the measure of the likelihood of an event occurring,

It ranges from 0 to 1, where 0 represents an impossible event and 1 represents a certain event.

Probability includes lot of concepts concepts among all those we are going to learn.

- Events and their relationship
  - Independent events
  - Dependent events,
- Conditional probability
- Counting / Combinatorics
  - Permutations
  - Combinations

Events and their relationship:

In **probability**, an **event** is a **specific outcome or a collection of possible outcomes** from a random experiment.

- Independent events
- Dependent events,

Independent Events

Two events are considered **independent** if the occurrence or non-occurrence of one event does not affect the probability of the other event occurring.

In other words, the events are independent if knowing the outcome of one event does not change the probability of the other event.

1) Toss Coin
2) Rolling Dice

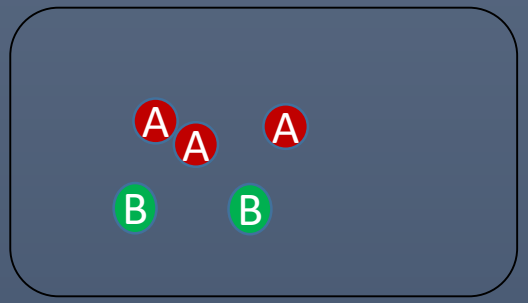
Rolling A dice

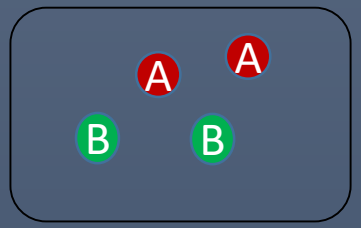


$$\Pr(1) = \frac{1}{6} \qquad \Pr(2) = \frac{2}{6} \qquad \Pr(3) = \frac{3}{6}$$

Dependent Events:

Two events are considered **dependent** if the occurrence of one event affects the probability of the other event occurring. In other words, the events are dependent if the probability of one event is influenced by the outcome of the other event.





Where:
P(A) = probability of A
P(B) = probability of B

P(B/A) = probability of B **given that A has already occurred**
P(A/B) = probability of A **given that B has already occurred**

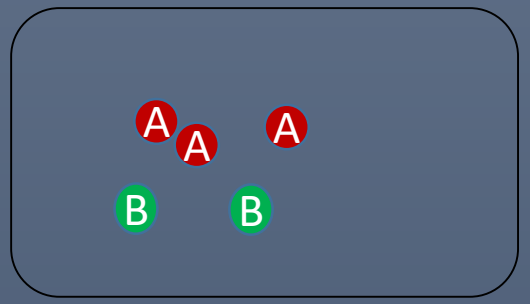
P(A∩B) = probability that **both A and B occur**

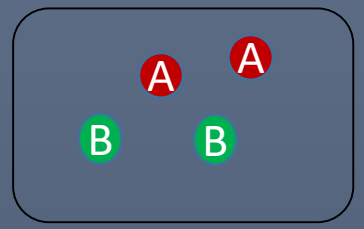
$Pr(A)\frac{3}{5}$

$Pr(A)\frac{2}{4}$

$Pr(B)\frac{2}{5}$

$Pr(B)\frac{2}{4}$





$Pr(A)\frac{3}{5}$

$Pr(A)\frac{2}{4}$

$Pr(B)\frac{2}{5}$

$Pr(B)\frac{2}{4}$

$$p\left(\frac{B}{A}\right) = \frac{p(B) * p\left(\frac{A}{B}\right)}{P(A)}$$

Where:
P(B/A) = probability of B **given that A has already occurred**

$$p\left(\frac{B}{A}\right) = \frac{\left(\frac{3}{5}\right) * \left(\frac{2}{4}\right)}{\frac{3}{5}}$$

$$= \frac{\left(\frac{3}{5} \frac{X}{X} \frac{2}{4}\right)}{\frac{3}{5}}$$

$$= \frac{\left(\frac{6}{20}\right)}{\frac{3}{5}}$$

$$= \frac{\left(\frac{3}{10}\right)}{\frac{3}{5}}$$

$$= \frac{\left(\frac{3}{10}\right)}{\frac{3}{5}}$$

$$= \frac{\left(\frac{3}{10}\right)}{\frac{3}{5}} = \frac{3}{10} \text{ X } \frac{5}{3} = \frac{15}{30} = \frac{1}{2}$$

$$p\left(\frac{B}{A}\right) = \frac{1}{2}$$

Counting / Combinatorics

It Focused on counting, arranging, and combining elements within sets according to specific rules.

- Permutations
- Combinations

Permutation:

A permutation is an arrangement of objects in a definite order. The members or elements of sets are arranged here in a sequence or linear order.

Example 01:

The permutation of set A={1,6} is 2, such as

- {1,6},
- {6,1}.

As you can see, there are no other ways to arrange the elements of set A.

Example 2:

The permutation of set B={1,4,5} is 6, such as

{1,4,5}
{1,5,4}

{4,5,1}
{4,1,5}

{5,1,4}
{5,4,1} .

Combination:

A combination is a mathematical technique that determines the number of possible arrangements in a collection of items where the **order of the selection does not matter with only unique combination** where items are **not repeating**.

Example01:

The Combination of set A={1,6} is 1, such as {1,6}. As you can see, there are no other ways to arrange the elements of set A with a unique combination where items are not repeating..

Example 2:

The permutation of set B={1,4,5} is 3, such as

As you can see, there are no other ways to arrange the elements of set B with unique combination where items are not repeating..